

BSDS (2025 Semester II): Statistics II

Quiz 1

NAME :	ROLL :
--------	--------

Time: 60 minutes

Total marks: 30

1. The number of COVID infected people in a certain community having N people is K_0 (K_0 is unknown and N is extremely large). Let $\theta = K_0/N$ be the proportion of infected people. To estimate θ , a person takes $n = 50$ simple random samples with replacement from this community.
 - (a) Employ an appropriate model to characterize this population. Justify your answer.
 - (b) Based on your assumed model, find the UMVUE of θ .

[5 + 5]

2. Let $X_1, \dots, X_n$ be a random sample from **Exponential**(θ) distribution, with common p.d.f.

$$f_X(x) = \theta \exp\{-\theta x\}, \quad x > 0, \quad \theta > 0.$$

Consider three estimates of $g(\theta) = \sqrt{\theta}$, $T_1(\mathbf{X}) = n^{-1} \sum_{i=1}^n X_i^{-1/2}$, and $T_2(\mathbf{X}) = (\bar{X}_n)^{-1/2}$.

- (a) Is $T_{1,n} = T_1(\mathbf{X})$ unbiased for $g(\theta)$? If not, can you find a function h_1 such that $h_1(T_{1,n})$ is unbiased for $g(\theta)$?

- (b) Is $T_{2,n} = T_2(\mathbf{X})$ unbiased for $g(\theta)$? If not, can you find a function h_2 such that $h_2(T_{2,n})$ is unbiased for $g(\theta)$? [Hint: You may use the additive property of Exponential distribution.]

- (c) Is any of the two unbiased estimators $h_1(T_{1,n})$ and $h_2(T_{2,n})$ the UMVUE of $\sqrt{\theta}$? Justify your answer. [4 + 5 + 5]

3. Let X be a random variable with parameters (θ, γ) p.d.f.

$$f_X(x) = \frac{\theta \gamma^\theta}{x^{\theta+1}}, \quad x \geq \gamma, \quad \theta > 0, \gamma > 0.$$

What is the distribution of $Y = \log X - \log \gamma$? Hence or otherwise find the expectation of $\log X$.
[4 + 2]